

MENGXU YUAN

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Education

University of Western Ontario
Ph.D. in Pure Mathematics

Sept. 2025 – Present
Supervisor: Prof. Hristo Sendov

University of Western Ontario
M.Sc. in Pure Mathematics

Sept. 2024 – Aug. 2025
Average: 88/100 Supervisor: Prof. Martin Pinsonnault

Wuhan University of Technology
B.Sc. in Information and Computing Science

Sept. 2020 – July 2024
Average: 90/100

Publications & Preprints

- [1] H. Sendov and M. Yuan. A Note on the De Bruijn-Springer Inequality between the Zeros and Critical Points of Polynomials. *Submitted*. PDF available at: mengxu3.github.io/assets/..., 2026.
- [2] H. Sendov and M. Yuan. Two Absolutely Bounded Determinantal Ratios. *arXiv preprint arXiv:2608.04294*, 2026.
- [3] H. Sendov and M. Yuan. Aggregate Bounds on the Eigenvalues of the Principal Submatrices of a Hermitian Matrix and Majorization Relations. *arXiv preprint arXiv:2601.16320*, 2026.
- [4] Y. Shi, W. Yang, M. Yuan, and Z. Yi. Few-Shot Font Generation Based on SAE and Diffusion Model. *Proc. AMMSAC*, Vol. 107, pp. 613–627, 2024.

Talks & Presentations

Aggregate Bounds on the Eigenvalues of the Principal Submatrices of a Hermitian Matrix and Majorization Relations
Speaker Feb. 2026

↪ Middlesex College 108, Western University

- Presented main results on aggregate bounds and majorization relationships based on the preprint arXiv:2601.16320.

Awards & Honors

Graduate Fellowship	University of Western Ontario	Sept. 2024 – Present
Western Graduate Research Scholarship	University of Western Ontario	Sept. 2024 – Present
Third-Class Scholarship	Wuhan University of Technology	Nov. 2023
Merit Student	Wuhan University of Technology	Nov. 2023
Second-Class Scholarship	Wuhan University of Technology	Nov. 2022
3 rd Prize, 14 th Huazhong Cup Math Modeling Competition	Hubei Society for Ind. and Appl. Math.	June 2022
1 st Prize, 13 th National College Student Mathematics Competition	Chinese Mathematical Society	Dec. 2021
National Endeavor Scholarship	Wuhan University of Technology	Nov. 2021
Merit Student	Wuhan University of Technology	Nov. 2021

Teaching Experience

University of Western Ontario Graduate Teaching Assistant
↪ Dept. of Mathematics

Sept. 2024 – Present

- **Summer 2026:** MATH 1600 (Linear Algebra I).
- **Winter 2026:** MATH 1225 (Methods of Calculus), AM 1201 (Calculus & Probability).
- **Fall 2025:** MATH 2155 (Math Structures), MATH 2151 (Discrete Math for Engineers).
- **Summer 2025:** MATH 1229 (Matrix Algebra), MATH 1228 (Finite Mathematics).
- **Winter 2025:** MATH 2700 (Linear Algebra II).
- **Fall 2024:** CALC 1000 (Calculus I).

Technical Skills

Programming Languages	Python, C/C++, MATLAB
Software & Tools	L ^A T _E X
Languages	Mandarin (Native), English (Fluent), Gan Chinese (Native)

Activities & Volunteer Service

English Conversation Program (ECP) *Participant* **Fall 2024**

↔ Western International, Western University

- Awarded the Certificate of Participation for engaging in cross-cultural communication and linguistic exchange.

Campus Talent Ambassador *Volunteer* **Aug. 2023 – Aug. 2025**

↔ Gongqingcheng City, Jiangxi, China

- Appointed by the Municipal Government to promote talent attraction policies to university students.

China Youth Innovation and Entrepreneurship Exchange Camp *Final Round Volunteer* **Sept. 2021**

↔ Hubei Provincial Volunteers Association

- Supported the final round of the 2021 8th *Creating Youth* China Youth Innovation and Entrepreneurship Competition (Technology Innovation Track).

Winter Vacation Alma Mater Trip **Enrollment Promotion** *Participant* **July 2021**

↔ Wuhan University of Technology

- Participated in the 2021 social practice and university enrollment promotion campaign.